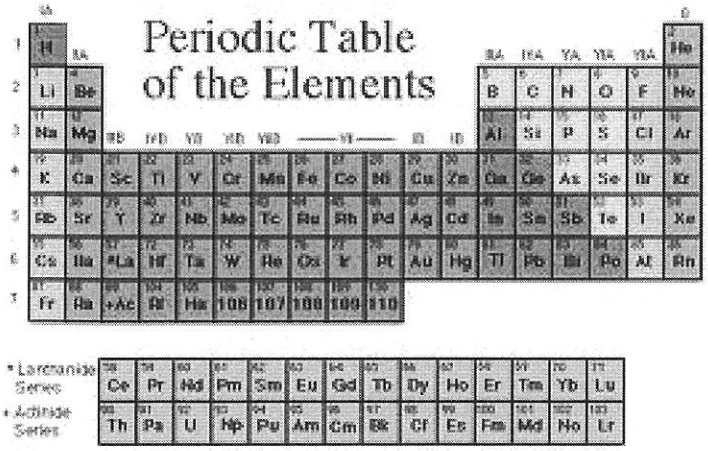
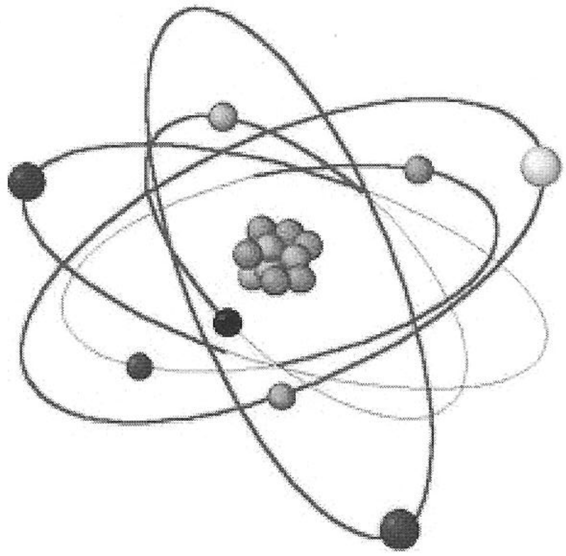
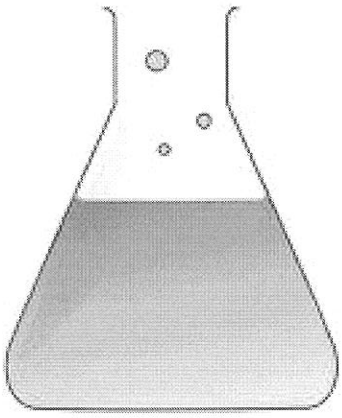


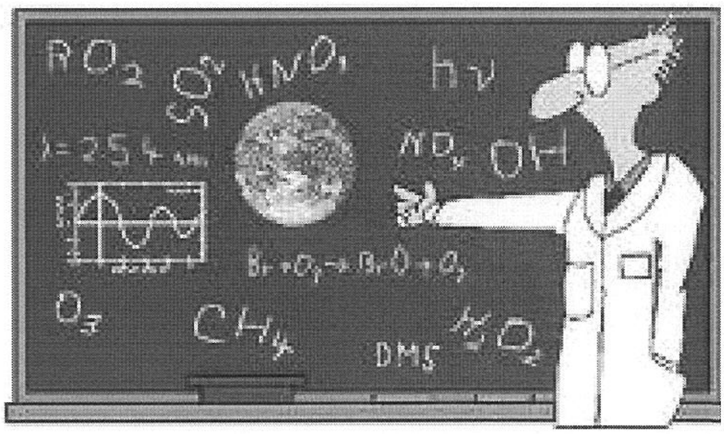


YEAR 9 SCIENCE
(Foundation)

CHEMISTRY

Name: ______________________________





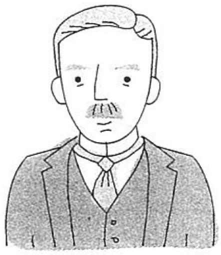
1.3 Model scientists

Science as a human endeavour

Visual/spatial **Verbal/linguistic**

Scientists have developed models to represent atoms. Refer to the Science as a Human Endeavour on pages 8–9 of your student book to read more about the development of atomic models.

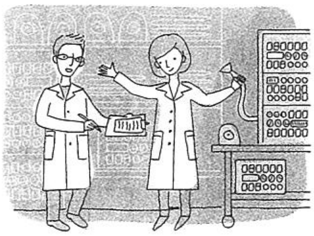
1 **Match** the scientist with the atomic model that made him famous. The first one has been done for you.



A Ernest Rutherford

B Neils Bohr

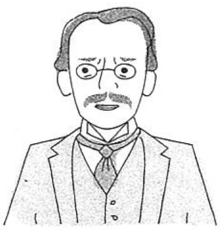
C Sir James Chadwick



D Modern atomic physicists

E Democritus

F Ancient Greek philosophers

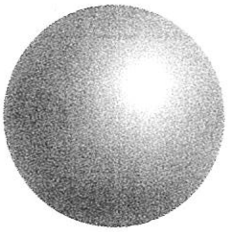


G Sir Joseph J. Thomson

H Philipp Lenard

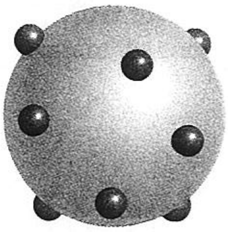
Continuum model
Substances are made up of the four elements: air, fire, water and earth. Substances can always be broken down into smaller pieces of the same substance.

Scientist: Ancient Greek philosophers



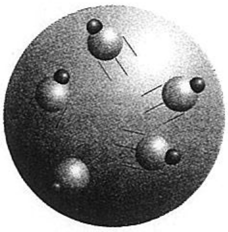
Hard sphere model
Substances are made up of hard, ball-like building blocks called atoms that cannot be broken apart.

Scientist: ______________________



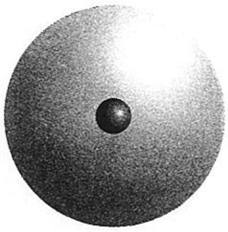
Plum pudding model
Atoms are made up of a positively charged ball with negatively charged electrons stuck in it.

Scientist: ______________________



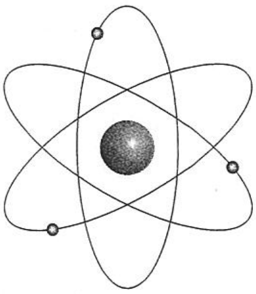
Dynamide model
Atoms are made up of moving particles called dynamides. Each dynamide is made up of a positively charged particle and a negatively charged particle.

Scientist: ______________________



Nuclear model
Atoms are made up of a solid, positively charged nucleus surrounded by an electron cloud.

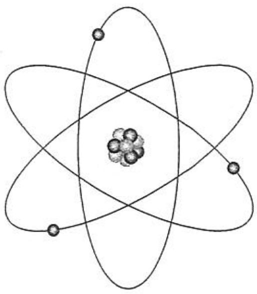
Scientist: ______________________



Planetary model
Atoms are made up of a solid, positively charged nucleus surrounded by electrons orbiting the nucleus, like planets orbiting the Sun.

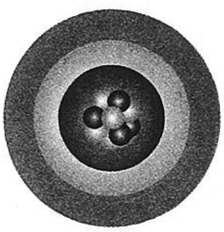
Scientist: ______________________

orbit (v) to move around something in a circle or ellipse



Neutron planetary model
Atoms are made up of a positive nucleus that has both protons and neutrons. The nucleus is surrounded by electrons orbiting it, like planets orbiting the Sun.

Scientist: ______________________



Electron shell model
Atoms are made up of a positive nucleus that has both protons and neutrons. The electrons form shells around the nucleus.

Scientist: ______________________

History of the atomic model

1a

1 Read the following text about scientists and cross out the incorrect verb alternatives. The verbs are underlined. The first one has been done for you. Remember that historical recounts should be written in past tense.

People often ~~made up~~ / make up models to explain / to explained things. Scientists perform / performs experiments which show / shows them how something acts, or what something looks / looked like. This information is used to develop / to develops a model. The model of the atom has underwent / has undergone many changes.

The idea of atoms is suggested / was suggested by Democritus in ancient Greece (460 – 370 BC). He logically deducted / deduced that matter could not continue to be cut into smaller pieces forever, and that eventually there had to be / have to be a smallest particle, which he calls / called the atom (about 400 BC).

In England, John Dalton (1766 – 1844) heard / heared about the experiments of other chemists. From their results he concluded / concludes that the atoms of the same element was / were identical and cannot be / could not be broken. In a chemical reaction, the atoms were separated, joined or rearranged. Dalton's model of the atom were / was like a ball (1803).

When English scientist Joseph Thomson (1856 – 1940), and other scientists, experiment / experimented with electricity, they finded / found that cathode rays are / were made up of tiny particles which travelled / travel in straight lines and carried / carryed a negative electric charge. These cathode rays came / comed from the negative terminal (ie the cathode) and travelled to the positive terminal (ie the anode), when very high voltages have / were applied. Thomson concludes / concluded that the rays make up / were made up of electrons, which were very small (they was/ were 2000 times lighter than a hydrogen atom) and had / have a negative charge. But if negative electrons came / come from neutral and relatively large atoms, the atom had to / has to look like a plum pudding: the dough of the pudding was a uniform substance with a positive charge and contained / contain tiny sultanas with a negative charge (1904).

2 In the following text about atoms and radioactivity, there are 12 spelling mistakes. Underline the incorrectly spelt words and write them correctly in the margin.

Radioactivity had been discovered and alpha particles were identified as tiny positive protons. A student of Joseph Thomson, the New Zealand born Ernest Rutherford (1871 – 1937), shot a beam of alpha particals at a very thin sheet of gold foil. He expected the protons to fly straight threw the foil, practically uneffected by the positive and negative particles in the gold atoms. Rutherford was suprised to find that some protons were deflected, some even bounced back to the sauce. Based on his measurements, Rutherford concluded that the atom's protons had to be concentrated in a small nucleus leaving the electrons to suround it. The mass of the atom was determined by the nucleus, which was much heavier than the electrons, and its size was determined by the electrons, which moved around the nucleus. Rutherford's model of an atom looked like a spherical cloud of negative electrons with a positive nucleus in the centre. If the atom was as big as a football feild, the nucleus would be as big as a tennis ball (1911).

The German Max Planck (1858 – 1947) formulated the quantum theory: Energy is given out or taken in by atoms only in lots (called quanta). Niels Bohr from Denmark (1885 – 1962) concluded that electrons jumped from one energy level to another, each jump being worth one qantum of energy. As a result the atomic model became like a soler system: a small nucleus, containing the heavy positive protons, was surrounded by several orbits on which the tiny negative electrons circled the nucleus (1913).

Erwin Schrodinger (1887 – 1961) and Wolfgang Pauli (1900 – 1958), both from Austria, proposed another model for the atom: The electron shells contained subshells of various shapes, each with a maximum of two electrons. But it was impossible to state where a particular electron was at any time; one could only say where an electron was most likley to be. Thus the atom appered like a nucleus inside many electron clouds (1926).

When the English scientist James Chadwick (1891 – 1974) discovered the existance of neutrons, which were as heavy as protons but carried no charge, the atomic model changed to include neutrons inside the nucleus together with the protons (1932).

History of the atomic model

3 The following words are snippets from sentences which summarise the text from questions 1 and 2. Sort the snippets from each sentence and then write down the sensible sentences.

a | atom | Democritus from ancient Greece | named | in about 400BC | the smallest particle |.

b | believed | in 1803 | like a ball | that | the atom | The English scientist John Dalton | was |.

c | drew | In 1904 | Joseph Thomson from England | like a positively charged plum pudding with negative plums in it | the atom |.

d | and | concentrated | Ernest Rutherford | In 1911 | in the central nucleus | moved | the negative electrons | onto a cloud surrounding the nucleus | the positive charge |, who was born in New Zealand, |.

e | but | in 1913 | kept | onto several orbits | placed | rather like our solar system | The Danish Niels Bohr | the negative electrons | the positive nucleus |.

f | an atom where the nucleus was surrounded by several orbital clouds | and | combined | developed | in 1926 | The Austrian scientists Erwin Schrodinger and Wolfgang Pauli | the idea of orbits and clouds |.

g | discovered | In 1932 | inside the positive nucleus | neutral neutrons | the English scientist James Chadwick |.

Drawing atomic models

In the paragraphs above various scientists developed certain models of atoms. Each model explained what the scientists knew at that time. New discoveries made new atomic models necessary.

4 Draw each model as described in the text. Next to your diagram write the description of the model as well as the name of the scientist and the year of discovery.

Atoms and molecules

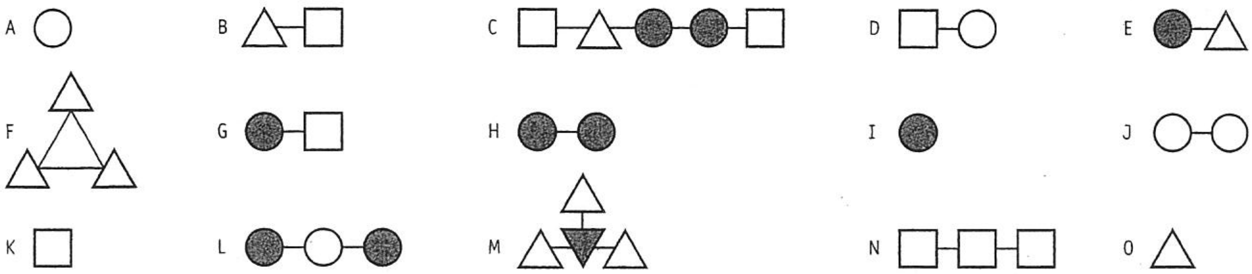
3

Matter can be divided into groups using certain criteria. Matter can be divided into living and non-living things. Matter can be divided into solids, liquids and gases. Matter can be grouped into metals and non-metals. Matter can be split into single particles and multiple particles. Matter can be divided into substances which contain one kind of atom or more than one kind of atom.

Definitions

- **Atoms** are tiny particles. They are the simplest form of matter. Atoms are made up of single particles.
- **Molecules** are also tiny particles. They are made up of two or more atoms. These atoms are held together by chemical bonds.
- **Elements** are substances which contain only one kind of atom.
- **Compounds** are substances which contain particles that are made up of two or more different atoms.

Look at the following diagrams showing various particles and substances. Then answer the questions below by referring to the definitions.



1 Identify all atoms and all molecules and record them in the table.

Particles	Atoms	Molecules
Diagrams		

2 Identify all elements and all compounds and record them in the table.

Substances	Elements	Compounds
Diagrams		

3 Complete the following table about particles and substances by adding as many examples of each as possible from the diagrams.

Atom of element	Atom of compound	Molecule of element	Molecule of compound

4 Explain why one cell in the table from question 3 is left empty. ____________________

____

5 Make up two more examples for each cell in question 3 and draw them below the table.

6 Explain what the lines in the diagrams represent. ____________________

____

Eureka! Solve this puzzle 4

Archimedes of Syracuse (287 – 212 BC) ran naked out into the streets crying 'Eureka!', excited after finding out the answer to King Hieron II's problem.

1 The following words relate to this topic about matter but are incomplete. Identify the missing letters and then use them and their matching numbers to complete the sentence below and discover the problem King Hieron had.

ma _ te _ (1, 6)	ato _ s (2)	s _ bstan _ e (3, 5)	mol _ _ ule _ (4, 5, 8)	_ a _ tic _ es (17, 6, 15)
_ ol _ d (8, 7)	b _ n _ s (11, 13)	_ or _ (9, 2)	_ riteri _ (5, 10)	m _ ta _ s (4, 15)
l _ qu _ d (7, 7)	s _ lid (11)	g _ s (10)	co _ poun _ s (2, 13)	_ ater (12)
sil _ e _ (14, 6)	_ ydr _ _ en (16, 11, 19)	_ le _ e _ t (4, 2, 18)		

King Hieron II's problem

_ _ _   _ _ _   _ _ _ _ _ _ _ _ _ _   _ _ _   _ _ _   _ _ _   _ _ _ _   _ _   _ _ _ _ _ _ _

13 7 13   1 16 4   19 11 15 13 8 2 7 1 16 8   3 8 4   10 15 15   1 16 4   19 11 15 13   11 6   6 4 17 15 10 5 4

_ _ _ _   _ _   _ _   _ _ _ _   _ _ _ _ _ _   _ _ _ _   _ _ k _ _ _   _ _ _   _ _ _ _ _ _   _ _ _ _ _?

8 11 2 4   11 9   7 1   12 7 1 16   8 7 15 14 4 6   12 16 4 18   2 10 k 7 18 19   1 16 4   19 11 15 13 4 18   5 6 11 12 18?

2 Write each of the words listed above into a sentence about science.

3 **Research:** Use library resources to find out how Archimedes solved the king's problem. This should explain why he was naked at the time!

ATOMIC AND MASS NUMBERS

Write in the charge on a

(i) proton ________

(ii) neutron ________

(iii) electron ________

Do you know the charge law?

Like charges ________

Unlike charges ________

Write down the name of the relationship *(attraction, repulsion)* between the following particles

proton and proton ________

proton and electron ________

proton and neutron ________

electron and electron ________

electron and neutron ________

The Atomic Number (Z) is the number of ________ in the ________
The Mass Number (A) is the number of ________ in the ________

Complete the following table

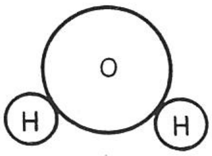
SUBSTANCE	NAME	MASS NUMBER (A)	ATOMIC NUMBER (Z) (No. OF PROTONS)	No. OF NEUTRONS (N)
$^{12}_{6}C$				
N		14		7
O			8	8
P		31	15	
$^{35}_{17}Cl$				
Ca		40		20
Fe				
Ag		108	47	

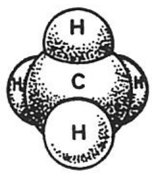
NAME THE COMPOUND

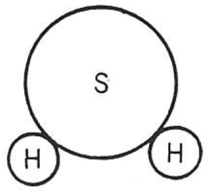
1. Complete this table.

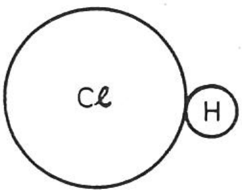
FORMULA OF SUBSTANCE	ELEMENTS CONTAINED IN SUBSTANCE	NAME OF SUBSTANCE
$CaCO_3$	calcium, carbon, oxygen	calcium carbonate
CO_2		
Al_2O_3		
KI		
Fe_2O_3		
SO_2		
NaCl		
$PbSO_4$		

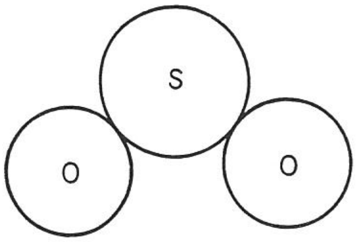
2. Examine these models of compounds and write the formula of each compound.

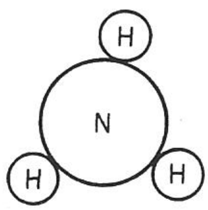


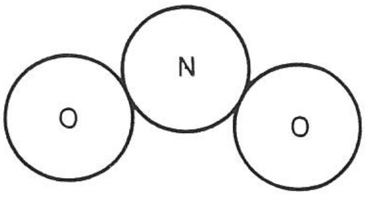


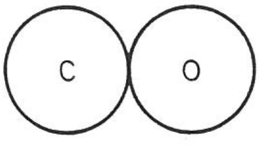


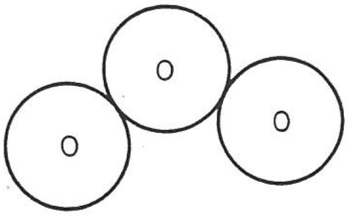












CHEMISTRY SHORTHAND

Scientists use a shorthand method of writing compounds. They use the chemical symbols of the *elements* that make them up. In addition, they tell us the *number of each type of atom* present in a *single molecule of the compound:*

The following table shows us some common examples of compounds that we see every day.

SUBSTANCE	FORMULA	EACH MOLECULE CONTAINS
WATER	H_2O	*2 atoms of Hydrogen, 1 atom of Oxygen*
SALT	NaCl	*1 atom of Sodium, 1 atom of Chlorine*
SUGAR	$C_{12}H_{22}O_{11}$	*12 atoms of Carbon, 22 atoms of Hydrogen, 11 atoms of Oxygen*
SAND	SiO_2	*1 atom of Silicon, 2 atoms of Oxygen*

QUESTIONS

1. Ammonia gas molecules consist of one atom of nitrogen and three atoms of hydrogen. What is its chemical formula?

   ____

2. Hydrogen sulfide is the poisonous 'rotten egg' gas. A molecule of this compound is composed of two atoms of hydrogen and one of sulfur. Write its formula.

   ____

3. If a nitrogen gas molecule consists of two atoms of nitrogen, what is its chemical formula?

   ____

4. Ozone gas molecules consist of three oxygen atoms. What is the chemical formula of ozone?

   ____

Complete the following table:-

COMPOUND	NUMBER AND TYPE OF ATOMS IN COMPOUND	FORMULA
Carbon dioxide	1 Carbon, 2 Oxygen	____________
Nitric acid	1 Hydrogen, 1 Nitrogen, 3 Oxygen	____________
Copper sulfate	________________________________	$CuSO_4$
Potassium permanganate	________________________________	$KMnO_4$
Potassium dichromate	________________________________	$K_2Cr_2O_7$
Sulfuric acid	2 Hydrogen, 1 Sulfur, 4 Oxygen	____________
Sodium hydrogen carbonate	________________________________	$NaHCO_3$
Copper carbonate	1 Copper, 1 Carbon, 3 Oxygen	____________
Acetic acid	________________________________	CH_3COOH
Sodium hydroxide	1 Sodium, 1 Oxygen, 1 Hydrogen	____________

ELEMENTS, COMPOUNDS AND MIXTURES

1. Elements are pure substances made up of only one kind of ____________
They cannot be broken down into simpler substances.

2. There are ______ naturally occurring elements and each has its own chemical symbol.

3. Write symbols for these elements:

aluminium ______	calcium ______	carbon ______	chlorine ______
copper ______	gold ______	hydrogen ______	iron ______
lead ______	magnesium ______	manganese ______	mercury ______
nickel ______	nitrogen ______	oxygen ______	silicon ______
silver ______	sodium ______	sulfur ______	zinc ______

4. Elements can react together chemically to form c ______________.

5. If substances are stirred or mixed together, without a chemical reaction taking place between them, then a m _________ is produced.

6. Indicate if the substances below are elements, compounds, or mixtures:

iron ____________________	carbon dioxide ____________________
air ____________________	water ____________________
salt solution ____________________	iron and sulfur ____________________
oxygen ____________________	sulfur ____________________

7. Chemists make use of prefixes in naming compounds. What do these prefixes mean?

mono _____, di _____, tri _____, tetra _____, penta _____

8. Write chemical formulas for these compounds:

carbon monoxide ____________	sulfur dioxide ____________
sulfur trioxide ____________	carbon tetrachloride ____________
phosphorus pentachloride _____	dinitrogen tetroxide ____________

9. Complete the table below:

FORMULA	NAME OF COMPOUND	NUMBER AND TYPE OF ATOM
CO_2		
NO_2		
$NaCl$		
CS_2		
HCl		
$CuSO_4$		
$CaCO_3$		

10. Write a word equation for the reaction between
A. iron and sulfur to form iron sulfide

____

B. magnesium and oxygen to form magnesium oxide

____

C. hydrogen and oxygen to form water

____

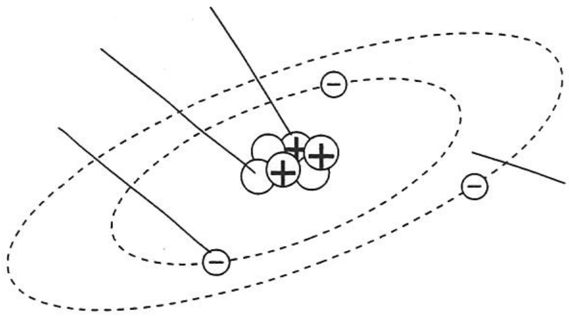
Atoms, molecules and ions

NAME ______________________ **CLASS** ______________

1 Use your ruler to draw straight lines matching each term on the left with its definition on the right.

atom	negatively charged particle that orbits around the nucleus of an atom
proton	pure substance made up of atoms of only one type
neutron	central part of an atom, containing protons and neutrons
electron	smallest piece of ordinary matter
nucleus	positively charged sub-atomic particle
element	organised chart of all known elements
molecule	chemical substance made up from two or more elements
compound	charged atom
ion	chemical substance made up from two or more atoms bonded together
periodic table	sub-atomic particle that has no charge

2 Label the model of an atom shown below.



This model represents an atom of ______________

Atoms, molecules and ions

NAME ______________________________ **CLASS** ______________

I	II											III	IV	V	VI	VII	VIII
1 H Hydrogen																	2 He Helium
3 Li Lithium	4 Be Beryllium											5 B Boron	6 C Carbon	7 N Nitrogen	8 O Oxygen	9 F Fluorine	10 Ne Neon
11 Na Sodium	12 Mg Magnesium											13 Al Aluminium	14 Si Silicon	15 P Phosphorus	16 S Sulfur	17 Cl Chlorine	18 Ar Argon
19 K Potassium	20 Ca Calcium	21 Sc Scandium	22 Ti Titanium	23 V Vanadium	24 Cr Chromium	25 Mn Manganese	26 Fe Iron	27 Co Cobalt	28 Ni Nickel	29 Cu Copper	30 Zn Zinc	31 Ga Gallium	32 Ge Germanium	33 As Arsenic	34 Se Selenium	35 Br Bromine	36 Kr Krypton
37 Rb Rubidium	38 Sr Strontium	39 Y Yttrium	40 Zr Zirconium	41 Nb Niobium	42 Mo Molybdenum	43 Tc Tantalum	44 Ru Tungsten	45 Rh Rhodium	46 Pd Palladium	47 Ag Silver	48 Cd Cadmium	49 In Indium	50 Sn Tin	51 Sb Antimony	52 Te Tellurium	53 I Iodine	54 Xe Xenon
55 Cs Cesium	56 Ba Barium	57 La Lanthanum	72 Hf Hafnium	73 Ta Tantalum	74 W Tungsten	75 Re Rhenium	76 Os Osmium	77 Ir Iridium	78 Pt Platinum	79 Au Gold	80 Hg Mercury	81 Tl Thallium	82 Pb Lead	83 Bi Bismuth	84 Po Polonium	85 At Astatine	86 Rn Radon
87 Fr Francium	88 Ra Radium	89 Ac Actinium	104 Rf Rutherfordium	105 Ha Hahnium	106 Sg Seaborgium	107 Ns Neilsbohrium	108 Hs Hassium	109 Mt Meitnerium	110 Uun Ununnilium		112 ?						

58 Ce Cerium	59 Pr Praseodymium	60 Nd Neodymium	61 Pm Promethium	62 Sm Samarium	63 Eu Europium	64 Gd Gadolinium	65 Tb Terbium	66 Dy Dysprosium	67 Ho Holmium	68 Er Erbium	69 Tm Thulium	70 Yb Ytterbium	71 Lu Lutetium
90 Th Thorium	91 Pa Protactinium	92 U Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium

Atomic number — 12
Symbol — Mg
Magnesium

1 Use coloured pencils to shade the metals and non-metals in the periodic table above.

2 List the 'noble gases'. ______________________________

3 a What is the first element listed in the periodic table? ______________

b What is its atomic number? ______________

4 What does the term 'atomic number' mean? ______________________

5 What is the chemical symbol for each of the following elements?

Helium ____________ Carbon ____________ Sodium ____________

Chlorine ____________ Sulfur ____________ Calcium ____________

6 a How many naturally occurring elements are there? ______________

b How many more have been manufactured artificially? ______________

7 What is the common feature of all of the elements in the first column of the periodic table? ______________________________

8 a How many protons are there in the nucleus of an atom of hydrogen? __________

b How many electrons are there in a neutral hydrogen atom? __________

ACTIVITY 5.4

Find the elements

Name:

Skills: literacy, knowledge

The names of 22 elements are hidden in this grid. The names go across, down and diagonally and can go backwards or forwards. The elements are from the two lists of elements below the grid.

In the second table the elements are shown only with their symbols. Find the correct spelling of each name, write it in the second table, then find it on the grid. You can use the periodic table from the next activity, Activity 5.5, to find the spelling.

M	U	I	S	E	N	G	A	M	O	O	S
A	Z	A	N	U	B	R	A	C	N	U	O
N	S	D	E	A	L	I	U	T	U	N	D
G	C	A	L	C	I	F	C	F	O	G	I
A	S	E	L	I	C	O	N	R	L	Y	E
N	A	L	U	M	I	N	I	U	M	U	M
E	M	Z	G	K	N	I	Z	U	K	K	S
S	M	U	I	D	O	S	I	L	V	E	R
E	N	O	B	R	A	C	A	V	L	I	S
N	I	T	R	O	L	C	O	P	P	E	R
D	T	U	N	A	S	T	E	N	U	Z	L
E	R	G	C	S	U	L	F	A	N	M	V
N	O	L	H	Y	D	R	O	G	E	E	E
E	G	O	L	X	C	N	X	O	G	R	R
T	E	D	O	A	L	O	Y	L	O	C	S
S	N	E	R	X	E	C	G	D	R	U	U
G	E	N	I	T	K	I	E	Y	D	R	L
N	U	U	N	R	C	L	N	I	Y	Y	V
U	E	O	E	U	I	I	E	G	H	N	E
T	I	I	N	E	N	S	U	L	F	E	R

Aluminium	Carbon	Magnesium	Mercury	Nitrogen	Silicon	Tungsten
Calcium	Hydrogen	Manganese	Nickel	Oxygen	Sulfur	Zinc

Symbol	Name of element
Cl	
Cu	
Au	
Fe	
Pb	
Ag	
Na	
Sn	

The periodic table—page 1

Skills: knowledge, interpreting, analysing

Key: H — chemical symbol; hydrogen — name of element; 1 — atomic number

	Group I alkaline metals	Group II alkaline earth metals	transition metals										Group III (nonmetals)	Group IV	Group V	Group VI	Group VII	Group VIII noble gases
Period 1	H hydrogen 1																	He helium 2
Period 2	Li lithium 3	Be beryllium 4											B boron 5	C carbon 6	N nitrogen 7	O oxygen 8	F fluorine 9	Ne neon 10
Period 3	Na sodium 11	Mg magnesium 12											Al aluminium 13	Si silicon 14	P phosphorus 15	S sulfur 16	Cl chlorine 17	Ar argon 18
Period 4	K potassium 19	Ca calcium 20	Sc scandium 21	Ti titanium 22	V vanadium 23	Cr chromium 24	Mn manganese 25	Fe iron 26	Co cobalt 27	Ni nickel 28	Cu copper 29	Zn zinc 30	Ga gallium 31	Ge germanium 32	As arsenic 33	Se selenium 34	Br bromine 35	Kr krypton 36
Period 5	Rb rubidium 37	Sr strontium 38	Y yttrium 39	Zr zirconium 40	Nb niobium 41	Mo molybdenum 42	Tc technetium 43	Ru ruthenium 44	Rh rhodium 45	Pd palladium 46	Ag silver 47	Cd cadmium 48	In indium 49	Sn tin 50	Sb antimony 51	Te tellurium 52	I iodine 53	Xe xenon 54
Period 6	Cs cesium 55	Ba barium 56	La* lanthanum 57	Hf hafnium 72	Ta tantalum 73	W tungsten 74	Re rhenium 75	Os osmium 76	Ir iridium 77	Pt platinum 78	Au gold 79	Hg mercury 80	Tl thallium 81	Pb lead 82	Bi bismuth 83	Po polonium 84	At astatine 85	Rn radon 86
Period 7	Fr francium 87	Ra radium 88	Ac actinium** 89	Rf rutherfordium 104	Db dubnium 105	Sg seaborgium 106	Bh bohrium 107	Hs hassium 108	Mt meitnerium 109	Ds darmstadtium 110	Rg roentgenium 111	Uub ununbium 112						

*Lanthanides 58–71	Ce cerium 58	Pr praseodymium 59	Nd neodymium 60	Pm promethium 61	Sm samarium 62	Eu europium 63	Gd gadolinium 64	Tb terbium 65	Dy dysprosium 66	Ho holmium 67	Er erbium 68	Tm thulium 69	Yb ytterbium 70	Lu lutetium 71
**Actinides 90–103	Th thorium 90	Pa protactinium 91	U uranium 92	Np neptunium 93	Pu plutonium 94	Am americium 95	Cm curium 96	Bk berkelium 97	Cf californium 98	Es einsteinium 99	Fm fermium 100	Md mendelevium 101	No nobelium 102	Lr lawrencium 103

The periodic table—page 2

Name:

The elements are organised in a grid called the periodic table. They are listed in order of their atomic number. They are also arranged so that elements that are alike appear near each other. The elements in a column (called a group) all have similar properties.

Questions

Use the periodic table to answer the following questions.

1 State the number of elements in total that are listed in this version of the periodic table.__________

2 State the atomic number and symbol for the following elements.

- lead __________
- gold __________
- oxygen __________
- potassium __________
- sulfur __________
- carbon __________
- silver __________
- iron __________
- sodium __________

3 State the name and symbol for atoms with the following atomic numbers:

- 9 __________
- 4 __________
- 40 __________
- 13 __________
- 92 __________
- 80 __________
- 22 __________
- 27 __________
- 15 __________
- 14 __________

4 a Record the names of all elements that start with the letter S.

b State the symbols for the elements you have chosen.

c Explain why some elements have one letter for their symbol and others have two.

5 a Some columns (or groups) have names given to them. List the names of the three groups that are labelled.

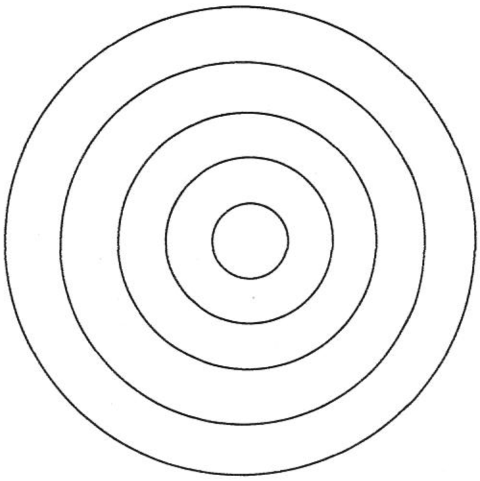
b In the group called the noble gases, list the name and symbol of the first three elements in the group.

6 Choose any two elements and describe some of their properties.

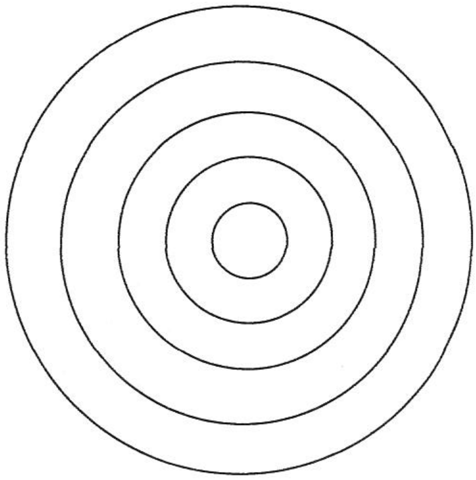
YEAR 9 SCIENCE
NATURAL AND PROCESSED MATERIALS
WORKSHEET - ELECTRON CONFIGURATION DRAWINGS

Draw electron configuration diagrams for the following elements. Show the ***number of protons and neutrons*** in the nucleus, and the ***number of electrons*** in each shell.

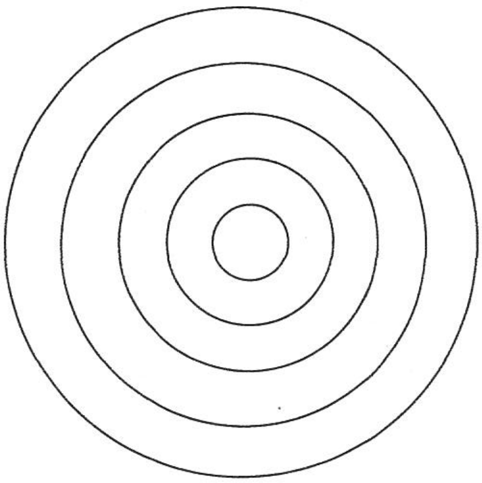
(a) $^{20}_{10}Ne$



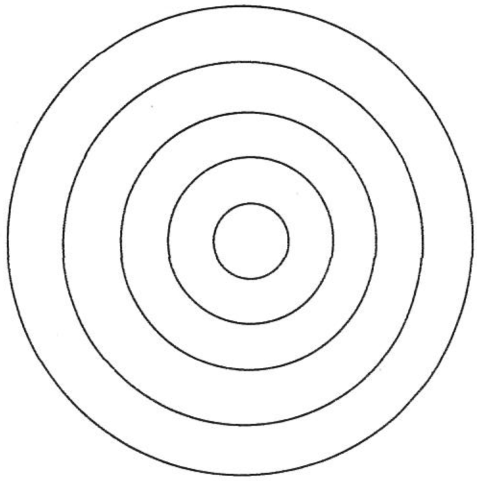
(b) $^{7}_{3}Li$



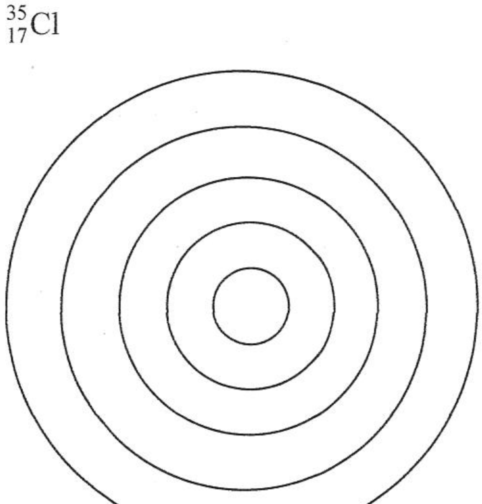
(c) $^{39}_{19}K$



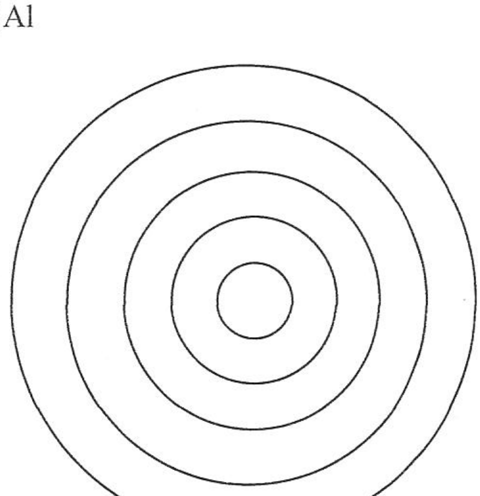
(d) $^{14}_{7}N$



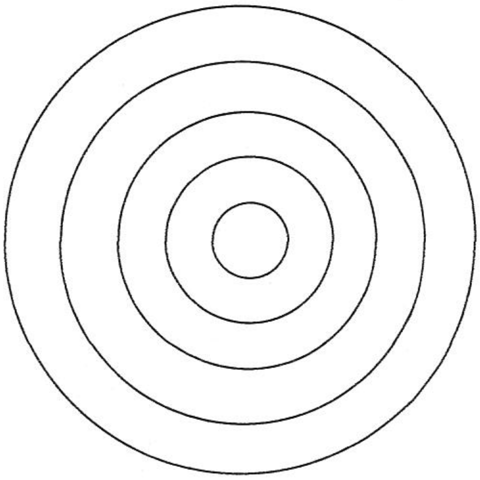
(e) $^{35}_{17}Cl$



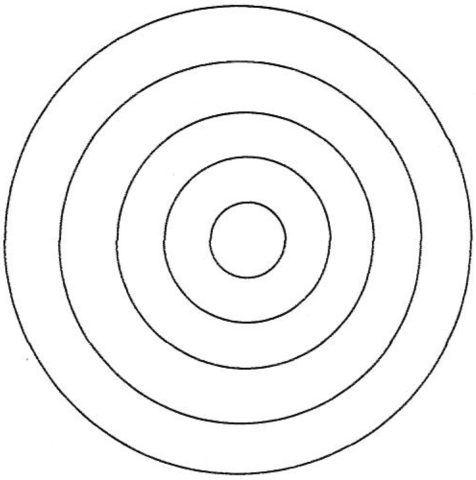
(f) $^{27}_{13}Al$



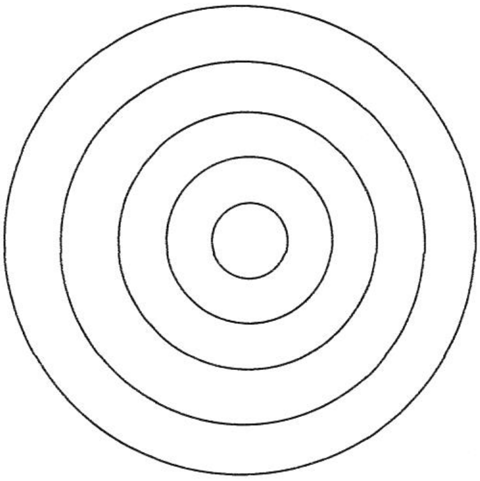
(g) $^{12}_{6}C$



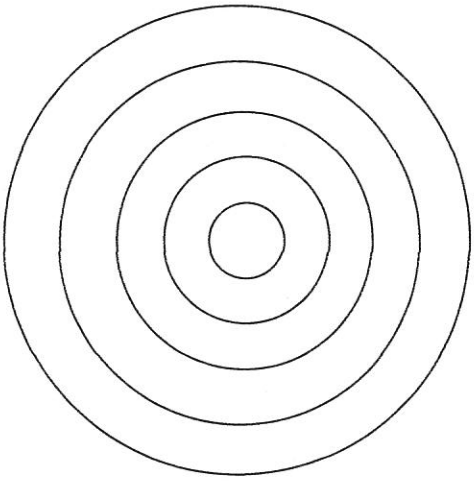
(h) $^{4}_{2}He$



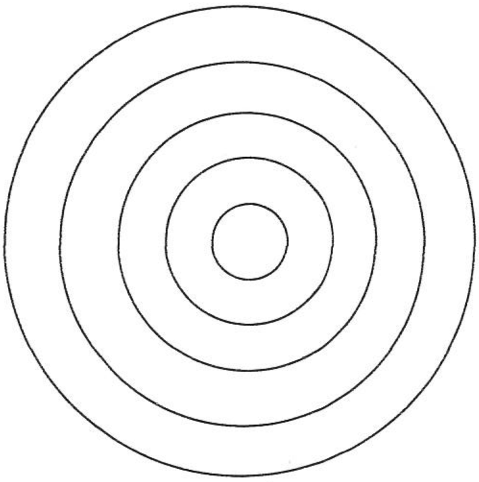
(i) $^{32}_{16}S$



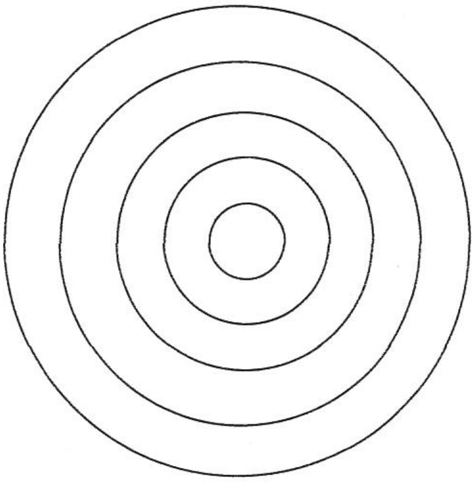
(j) $^{31}_{15}P$



(k) $^{10}_{5}B$



(l) $^{40}_{20}Ca$



Atomic structure

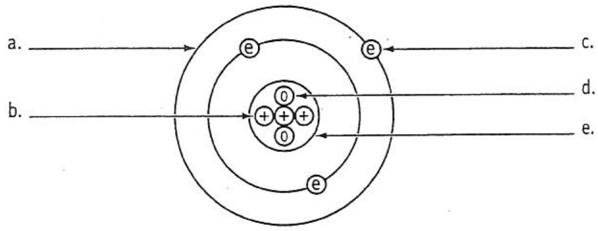
8a

Note: You will need to refer to a copy of the periodic table for this worksheet.

A popular atomic model

Many people today believe that atoms can be described by the atomic model of the solar system atom. This orbital atomic model allows us to explain many processes.

The atom is made up of three main subatomic particles which are grouped into two lots. Protons and neutrons are tightly packed into the nucleus which forms the centre of the atom. Electrons surround the nucleus on several orbits (or shells).



1 Label the diagram of an atomic model (lithium).

Subatomic particles

Protons and neutrons are 2000 times heavier than electrons. Protons and electrons have an electric charge: protons are positive and electrons are negative. Electrons fly around the nucleus on orbits. Neutrons and protons rest in the nucleus.

2 Complete the table about subatomic particles.

Name	Charge	Mass	Position	Movement
			on orbits	
		2000 times		
	neutral			no

Atomic numbers and mass numbers, atoms and isotopes

Atoms are neutral, although some of their subatomic particles do have an electric charge. But the positive and negative subatomic charges cancel each other leaving the atom neutral. Therefore, the number of positive protons must equal the number of negative electrons. The *atomic number* gives the number of protons.

Since protons and neutrons are very much heavier than electrons, it is the protons and neutrons which determine the mass of the atoms. Usually an atom has as many neutrons as protons. The *mass number* is the sum of protons and neutrons.

Some elements have atoms which have the same number of protons in the nucleus (so the same atomic number) but more or fewer neutrons (so they have different atomic weights). Their atoms are now called *isotopes*.

When the number of neutrons is much larger than the number of protons in a large nucleus, the isotope's nucleus is unstable. As a result it breaks up, forming new atoms and giving off radiation.

3 Complete the table about mass and atomic numbers of atoms.

Name	Mass number	Atomic number	Number of protons	Number of neutrons	Number of electrons	Scientific notation
lithium	5	3	3	2	3	$^{5}_{3}Li$
magnesium	24				12	
aluminium		13		13		
hydrogen	1					
chlorine						$^{35}_{17}Cl$
phosphorus						$^{31}_{15}P$
oxygen						$^{16}_{8}O$

Atomic structure

Electrons on shells

Electrons move on orbits (or shells) around the nucleus. The inside shells are smaller than the outer shells and so can hold fewer electrons. Inner shells are filled up before the outer shells. The first innermost shell can carry a maximum of two electrons, the second and third shells can hold no more than eight electrons (these maximum numbers are true for the first 20 atoms).

4 Draw atomic models for the following atoms. The mass and atomic numbers give you a clue to the number of protons, neutrons and electrons.

helium ($^{4}_{2}He$)

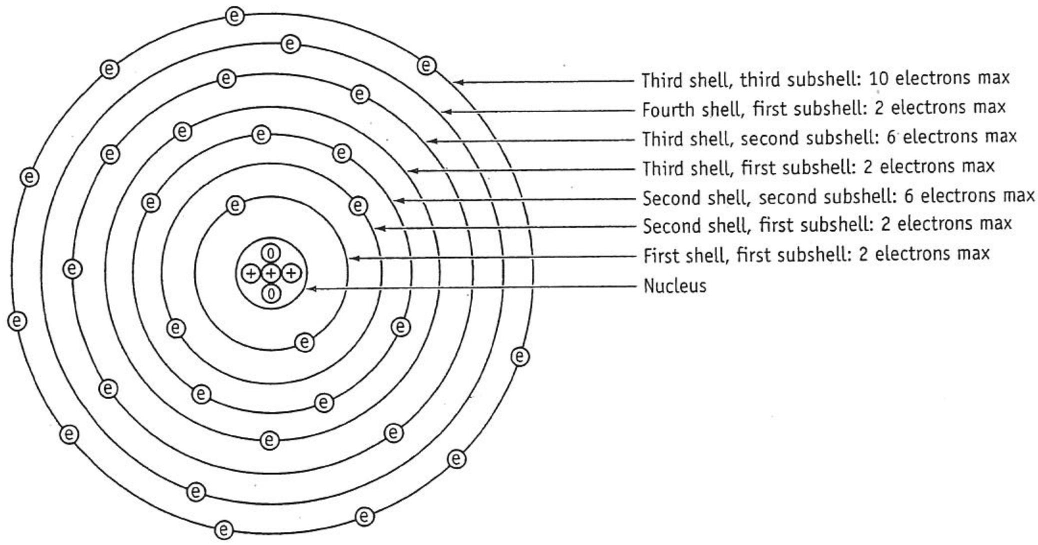
silicon ($^{28}_{14}Si$)

beryllium ($^{9}_{4}Be$)

potassium ($^{39}_{19}K$)

Shells and their subshells

The orbital model of the atom becomes untidy with the twenty-first atom, because the electron shells are actually composed of up to four subshells. And some of the subshells overlap each other. The third subshell of the third shell is actually bigger than the first subshell of the fourth shell.



5 State the number of protons, neutrons and electrons for the atom of iron ($^{56}_{26}Fe$).

____

6 Shade in electrons to the atomic model above for iron.

7 State the number of subshells for the first three shells. ______________________________